FPT SCHOOL OF BUSINESS & TECHNOLOGY

AI - MACHINE LEARNING

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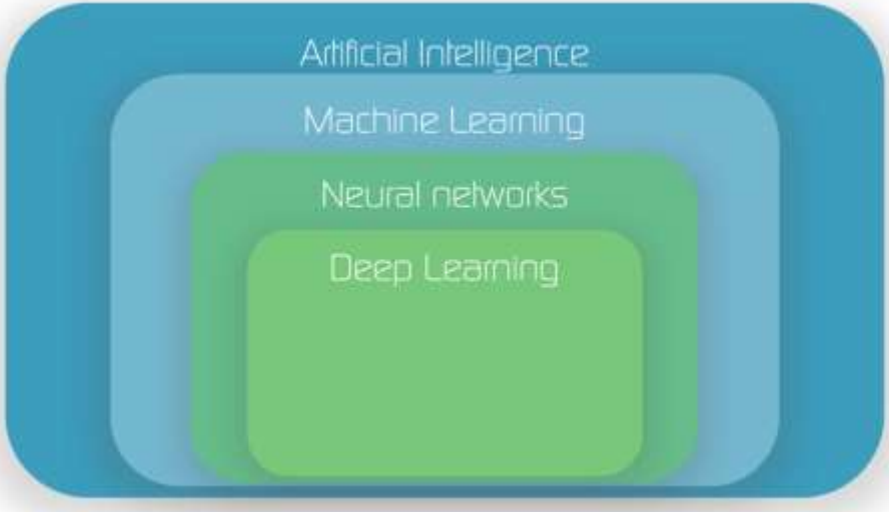
ARTIFICIAL INTELLIGENCE



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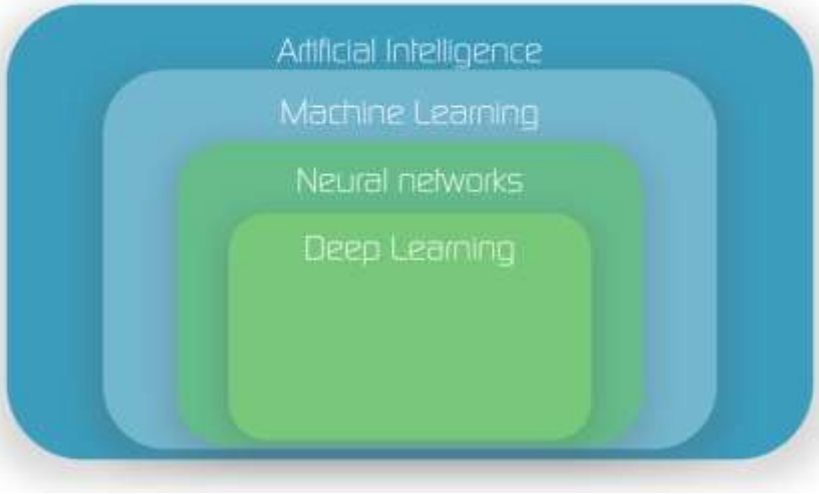


https://www.stoodnt.com/blog/ann-neural-networks-deep-learning-machine-learning-artificial-intelligence-differences/

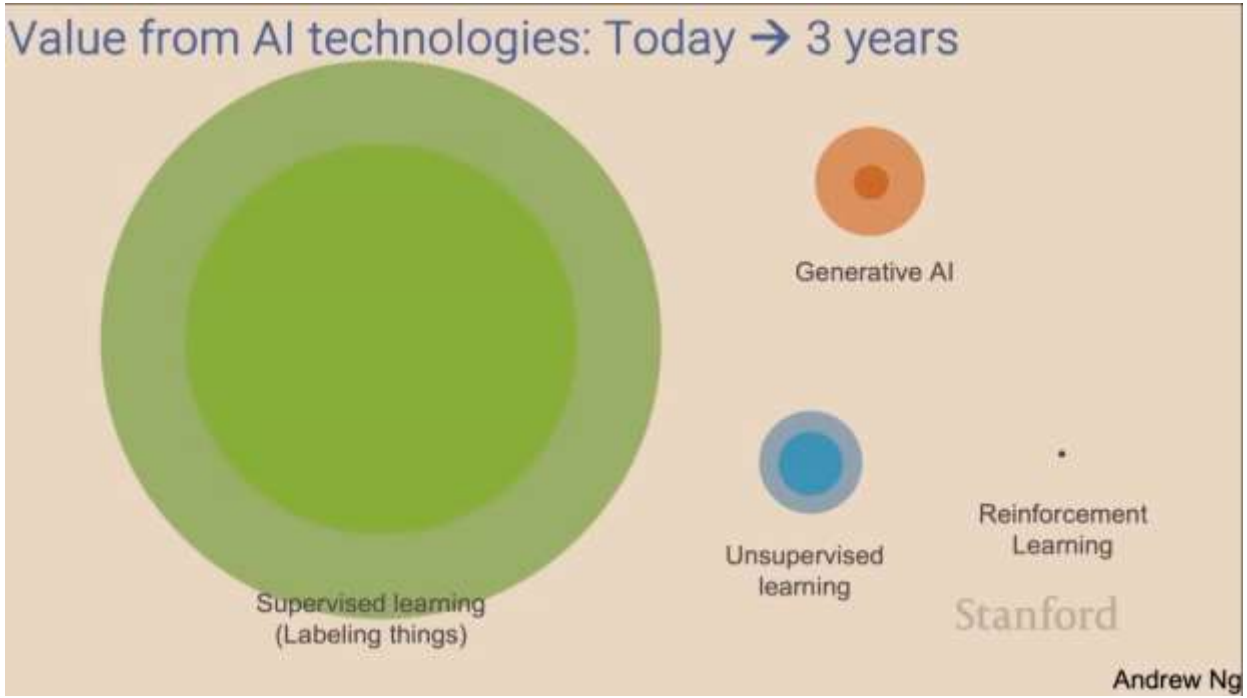
"science of making computers do things that require intelligence when done by humans"

"ability for a computer to output or does something that it wasn't programmed to do"

"a computer simulation of the way biological neurons work within a human brain"

"deep neural

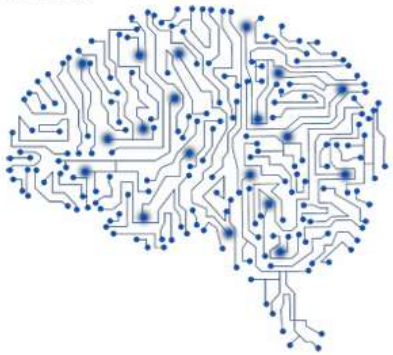
AI Opportunities



https://www.youtube.com/watch?v=5p248yoa3oE

What is Deep Learning?

- Deep Learning is an artificial intelligence function that imitates the workings of the human brain in processing data and creating patterns for use in decision making.
- Deep Learning transformed internet businesses like web search and advertising.
- Deep Learning is enabling brand new products and businesses.



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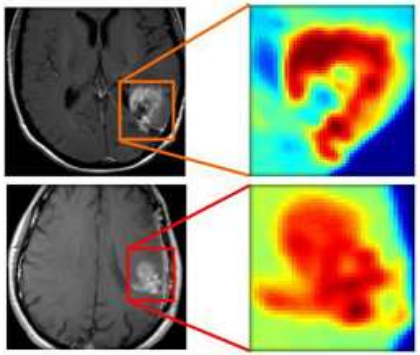
What is Deep Learning?

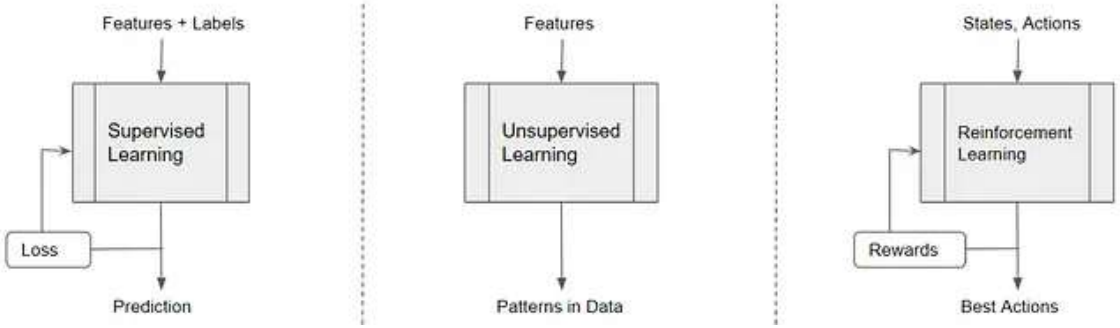
- Deep Learning uses:
  - Medicine (reading x-ray images)
  - Self driving car
  - Agriculture (precision agriculture)
  - And many more...



What is Deep Learning?

- Deep learning refers to training neural networks sometimes very large neural networks.
- A Neural Network, is a technology built to **simulate the activity of the human brain**. Specifically, pattern recognition and the passage of input through various layers of simulated neural connections.
- Inspired by human mind
- Neurons are **activated** by **some** input
- Massively used for problems of classification and regression
- Archived amazing results in the field
- Not so new, developed in 1940s, 1950s and 1960s

Machine Learning



- **Supervised learning - Học có giám sát**
  - Thuật toán ML cần học cách để chuyển đổi mỗi **input** (đầu vào) thành **label** (kết quả) tương ứng.
  - **Mọi mẫu dữ liệu huấn luyện** trong tập dữ liệu huấn luyện cần phải **biết trước label** của nó.
  - Bài toán phân loại (classification) và hồi quy (regression).
- **Unsupervised learning – Học ko giám sát**
  - Các mẫu dữ liệu trong học không giám sát **chỉ cần input** (đầu vào) mà **không cần label** (đầu ra).
  - Nó được sử dụng nhiều trong việc **khám phá cấu trúc** và **mối quan hệ của dữ liệu**.
  - Bài toán phân cụm (clustering): K-means, …
- **Reinforcement learning – Học tăng cường**
  - **Học hành vi tối ưu** trong một **môi trường** để đạt được **phần thưởng tối đa**.
  - Tự khám phá **chuỗi hành động** tối đa hóa phần thưởng.
  - Quá trình khám phá này giống như **tìm kiếm thử và sai**.

Classification vs Regression

CLASSIFICATION:

- A classification problem requires that examples be classified into one of two or more classes.
- A classification can have real-valued or discrete input variables.
- A problem with two classes is often called a two-class or binary classification problem.
- A problem with more than two classes is often called a multi-class classification problem.
- A problem where an example is assigned multiple classes is called a multi-label classification problem.

The classification accuracy is the percentage of correctly classified examples out of all predictions made.

REGRESSION:

- A regression problem requires the prediction of a quantity.
- A regression can have real valued or discrete input variables.
- A problem with multiple input variables is often called a multivariate regression problem.
- A regression problem where input variables are ordered by time is called a time series forecasting problem.

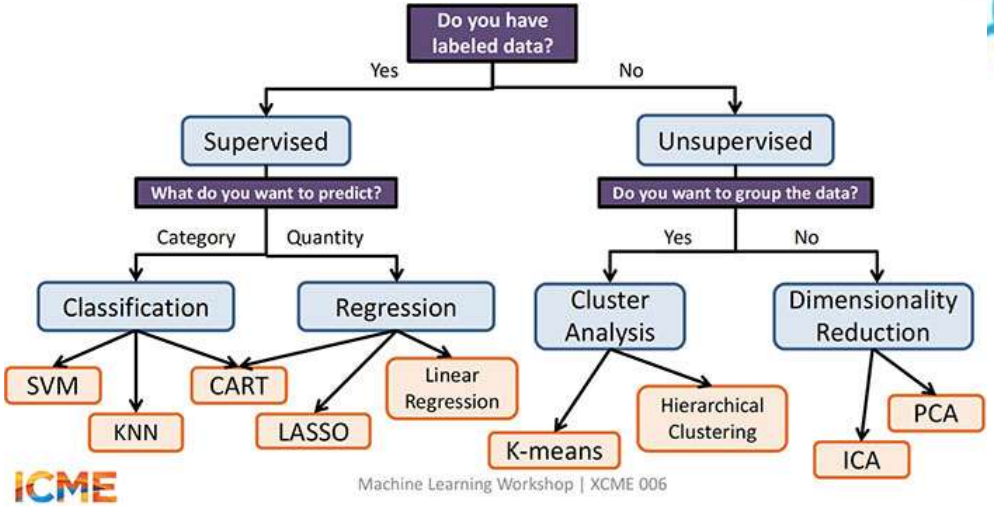
Most common estimation is to calculate the root mean squared error, abbreviated by the acronym RMSE

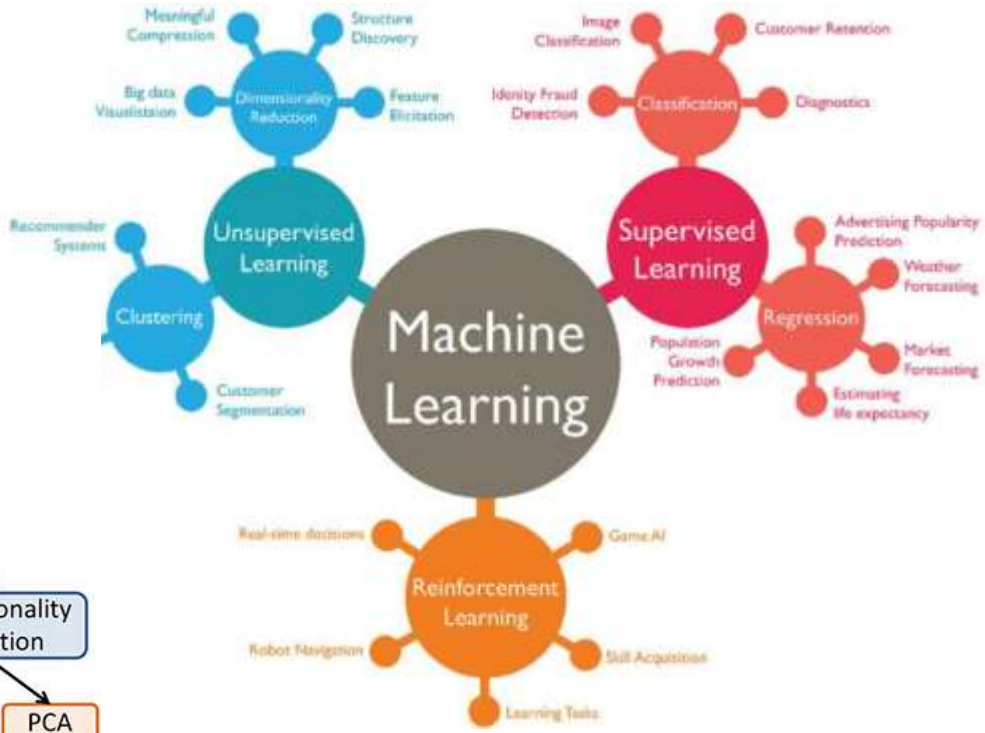
https://machinelearningmastery.com/classification-versus-regression-in-machine-learning/

Which Machine Learning Algorithm Should We Use?

- **Applications**

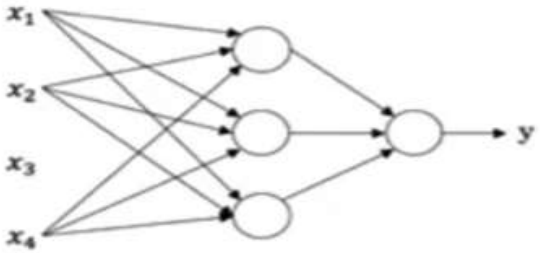
- **Categories**

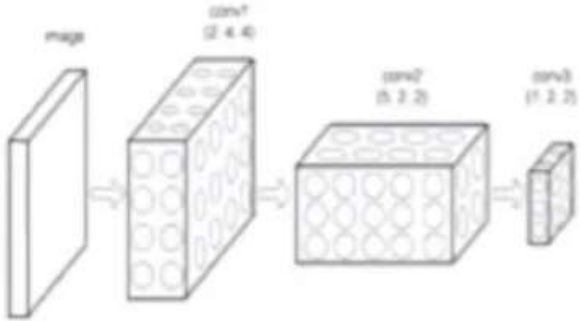




Course

- What will we learn?
  - Neural Networks and Deep Learning
    - You will learn how to build a neural network including a deep neural network
    - How to train it on data.
    - In the end, you will be able to build a deep neural network to recognize cats.
  - Improving Deep Neural Networks
    - Hyperparameter tuning, Regularization, Optimization, how to diagnose bias and variance.
  - Structuring your Machine Learning project
    - The strategy has changed in the era of deep learning.
    - A split of train and test
    - End-to-end deep learning
  - Convolutional Neural Networks
    - Often applied to images
  - Natural Language Processing
    - Building sequence models
    - Applying them to sequence data
    - Natural language is just a sequence of words





What is a NEURAL NETWORK?

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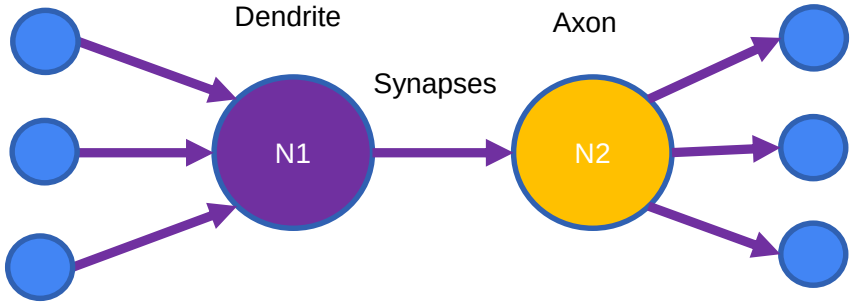
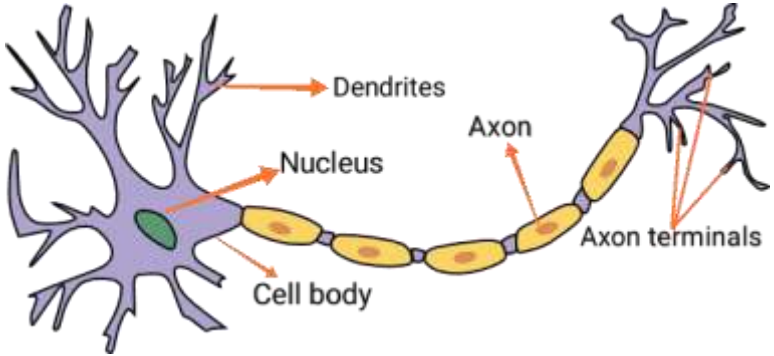
What is an Artificial Neural Network?





- Mạng nơ-ron nhân tạo (Neural Network) là một loại mô hình tính toán lấy cảm hứng từ cấu trúc của não bộ con người.
- Mỗi nơ-ron sinh học là một nút được kết nối với các nút khác thông qua các liên kết tương ứng với các kết nối sợi trục-khớp thần kinh-dendrite sinh học.
- Mỗi liên kết có một trọng số, xác định cường độ ảnh hưởng của một nút lên nút khác.

How Does A Baby Learn?

Input

Output

Cat

Cat

Dog

Pig

How Do Neural Networks Learn?

Input

Size

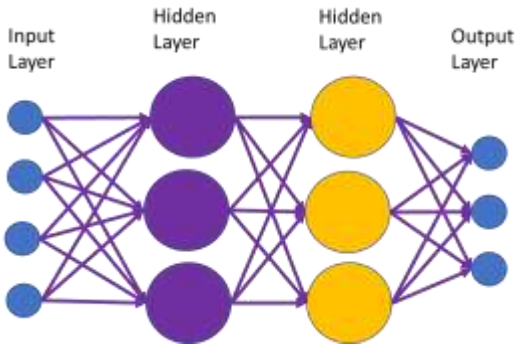
[32x26]

[32x32]

[32x42]

[32x28]

[32x32]



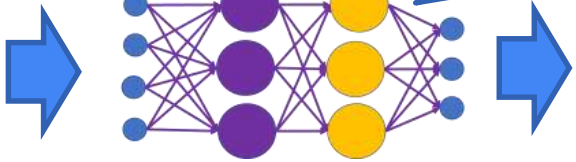
REAL NUMBER **32x32** **H1** **H2** **3**

Output

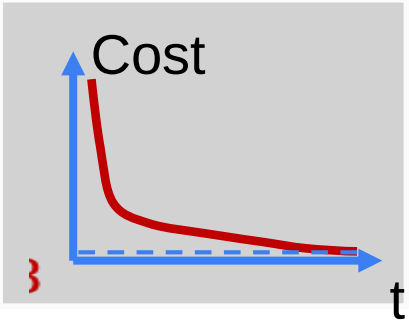
Actual	Target	
0.92	**1 Cat**	Cat
0.07		
0.05		
0.82	**1 Cat**	Cat
0.17		
0.15		
0. 35	**2 Dog**	Dog
1.67		
0.05		
0. 15	**3 Pig**	Pig
0.17		
2. 78		

How Do Neural Networks Learn?

Input



What is objective of the learning?



Loss= Target – Actual

$$\textbf{Cost} = \frac{1}{m}\sum_{i=0}^{m} Cost_i$$

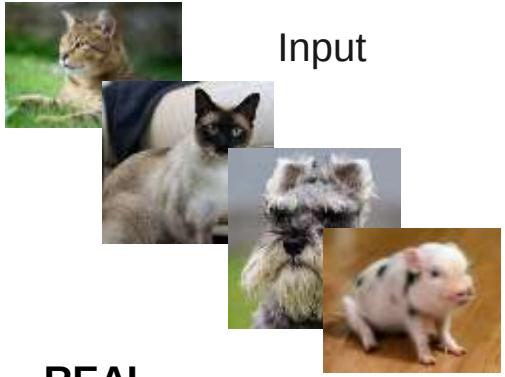
Output

Actual	Target
0.92	1 Cat
0.07	
0.05	
0.82	1 Cat
0.17	
0.15	
0. 35	2 Dog
1.67	
0.05	
0. 15	3 Pig
0.17	
2. 78	

REAL NUMBER

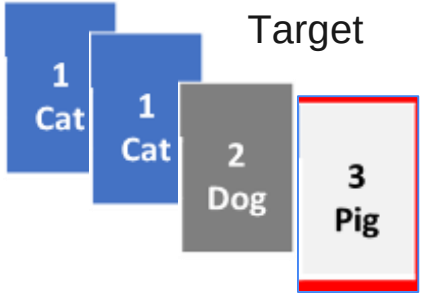
Note: Cost_i is a typo, the correct is Loss_i
Cost is about the whole training datasets, while Loss refers to individual item

How Do Neural Networks Learn?

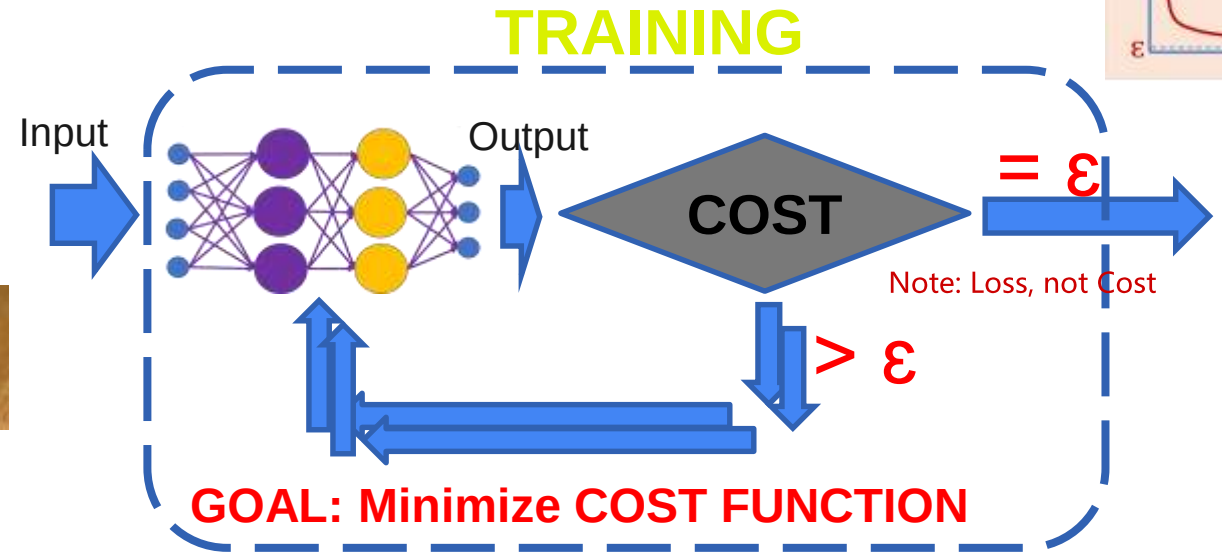


Input

REAL NUMBER



Target

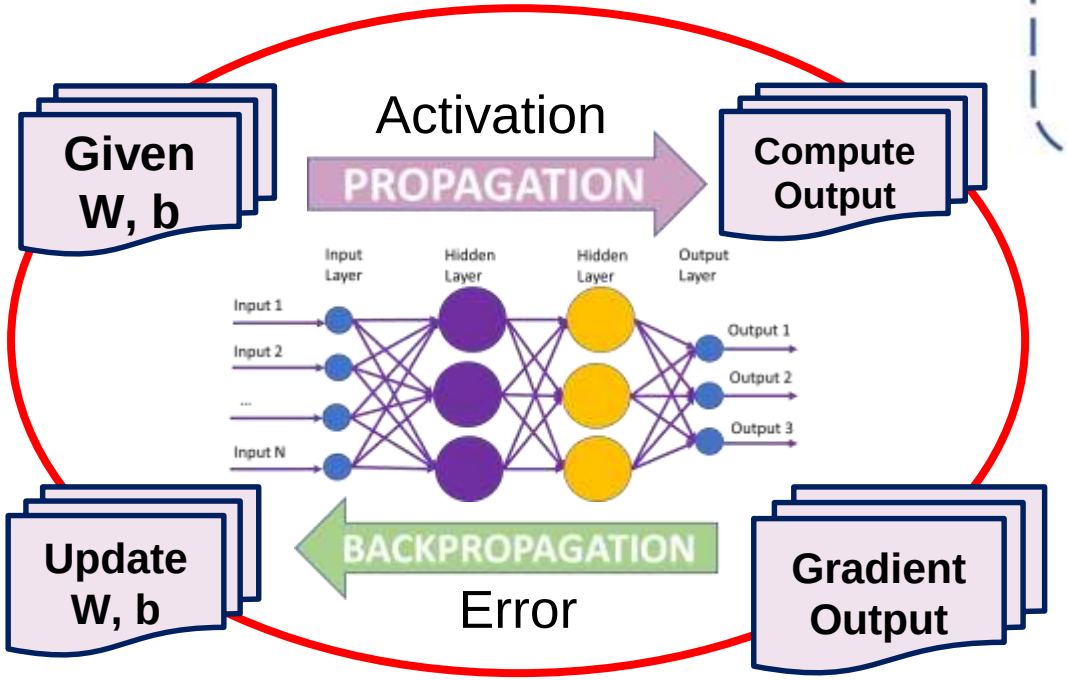


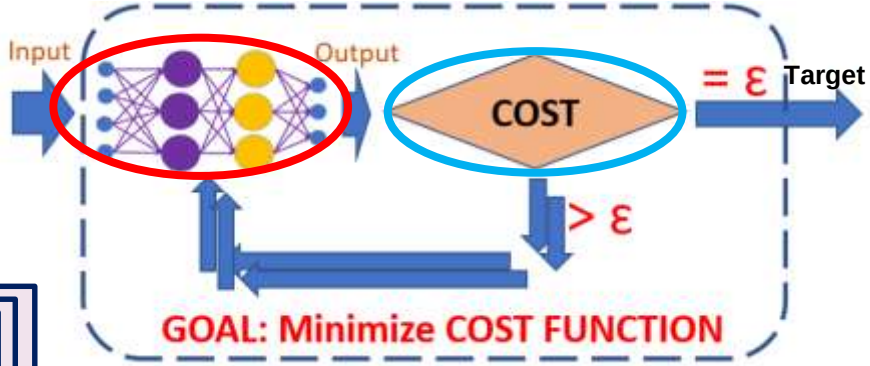
Các bước tiến hành huấn luyện model:

- Khởi tạo giá trị ban đầu cho model gồm trọng số weights và bias
- Tính toán "giá trị tổn thất" Cost
- Lặp lại quá trình training to **cập nhật model** khi **Cost > ε**
- Ngừng Training khi giá trị Loss bằng một giá trị nhỏ cho phép **Cost = ε**

Nguyên lý training của mạng nơ-ron?

SUPERVISED LEARNING





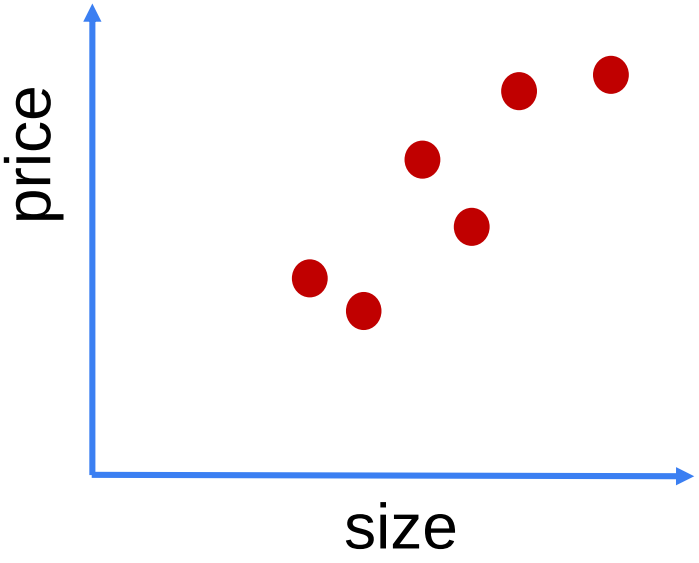
Loss = ***Target – Actual***

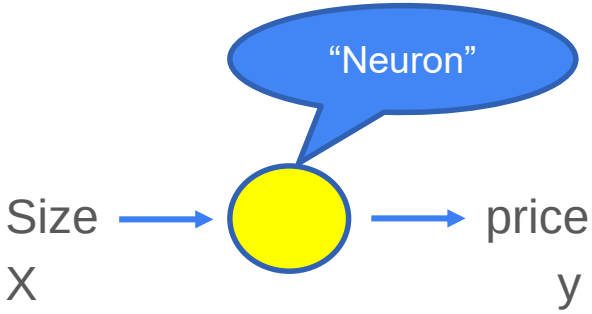
$$\textbf{Cost} = \frac{1}{m}\sum_{i=0}^{m} Loss_i$$

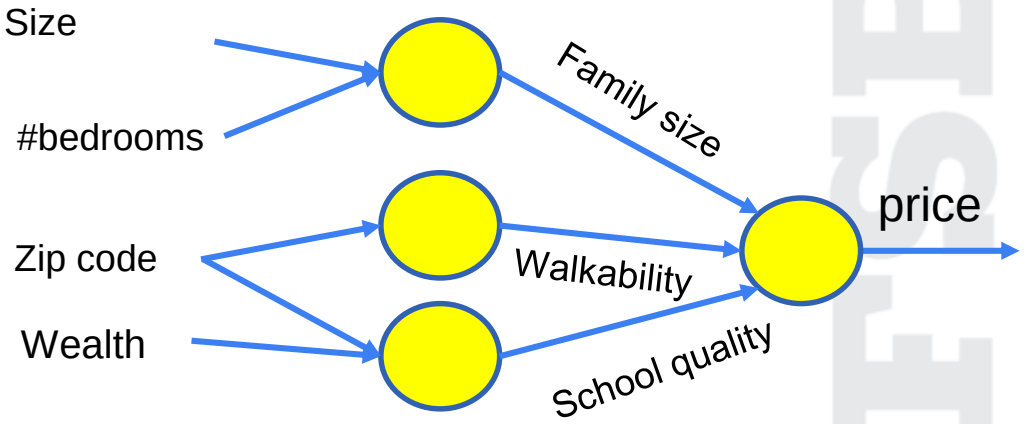
Where can neural networks be applied

- Speech Recognition
- Object Recognition on Images
- Driving Cars (Googles self-driving car)
- House Price Prediction
- Image Processing (Face Recognition)
- Advertisement (Whether a user would click on an advertisement {0, 1 })

House Price Prediction

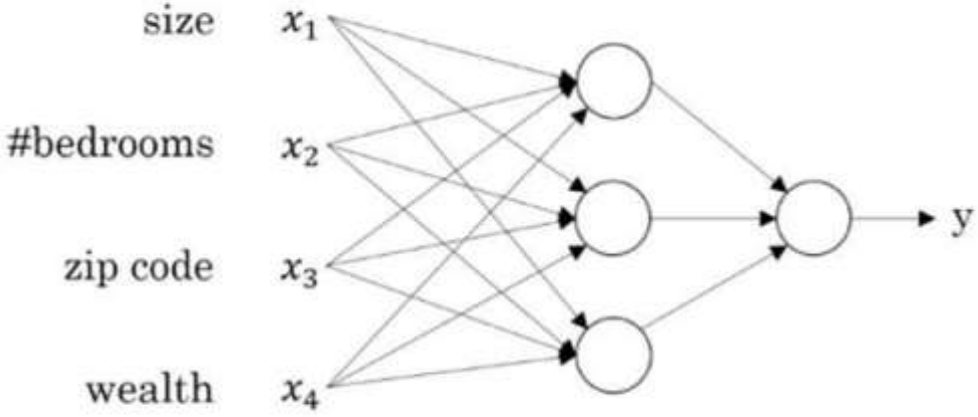






House Price Prediction

- In our example, " X " are all of these four inputs and " Y " is the price we are trying to predict. So by stacking together a few of the single neurons we now have a slightly larger network.
- The magic of a neural network is that when you implement it, you need to give it just the inputs " X " and the output " Y " for a number of examples in your training set and all these things in the middle, they will figure out by itself.

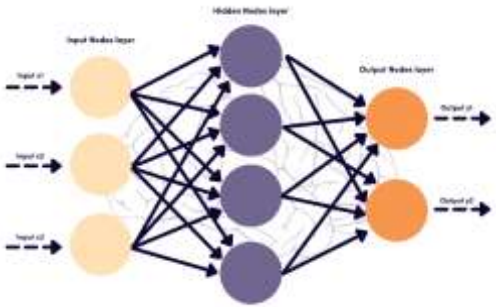


Supervised Learning

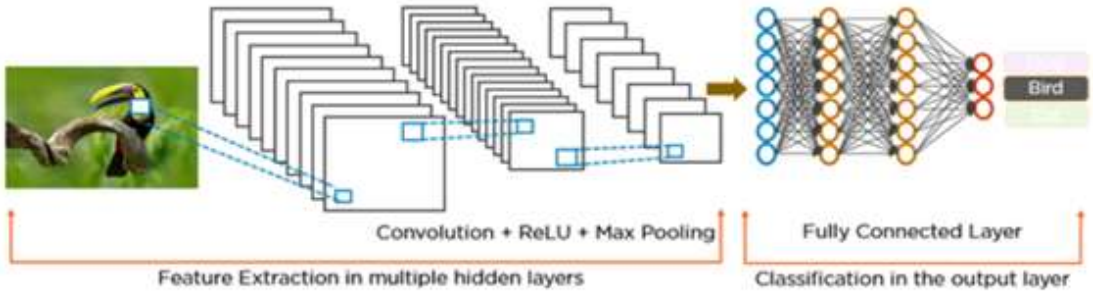
Input(x)	Output(y)	Application	
Home features	*Price*	***Real Estate***	Standard Neural Network
Ad, user info	*Click on ad?(0/1)*	***Online Advertising***	Standard Neural Network
Image	*Object (1,....,1000)*	***Photo tagging***	Convolutional Neural Network
Audio	*Text transcript*	***Speech recognition***	Recurrent Neural Network
English	*Chinese*	***Machine translation***	Recurrent Neural Network
Image, Radar info	*Position of other cars*	***Autonomous driving***	Custom / Hybrid Neural Network

Neural Network examples

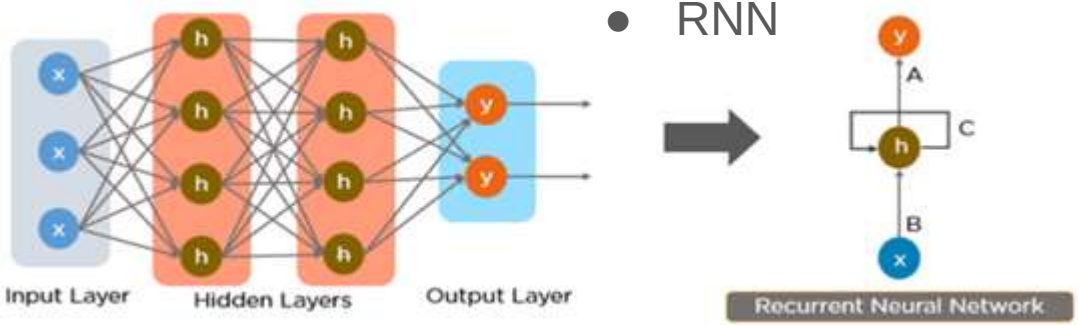
- ANN



- CNN



- RNN



Supervised Learning

Structured Data

- **databases** of data with a columns.
- each of the features such as size of a house on upper bedrooms or the age of a user has a **very well-defined meaning**.

Size	#bedrooms	Price(1000$s)
2104	3	400
1600	4	540

User Age	Ad Id	Click
41	93242	0
18	71244	1

Unstructured Data

- refers to things like **raw audio, images or text**.
- it's been much harder for computers to make sense of unstructured data compared to structured data.

Audio

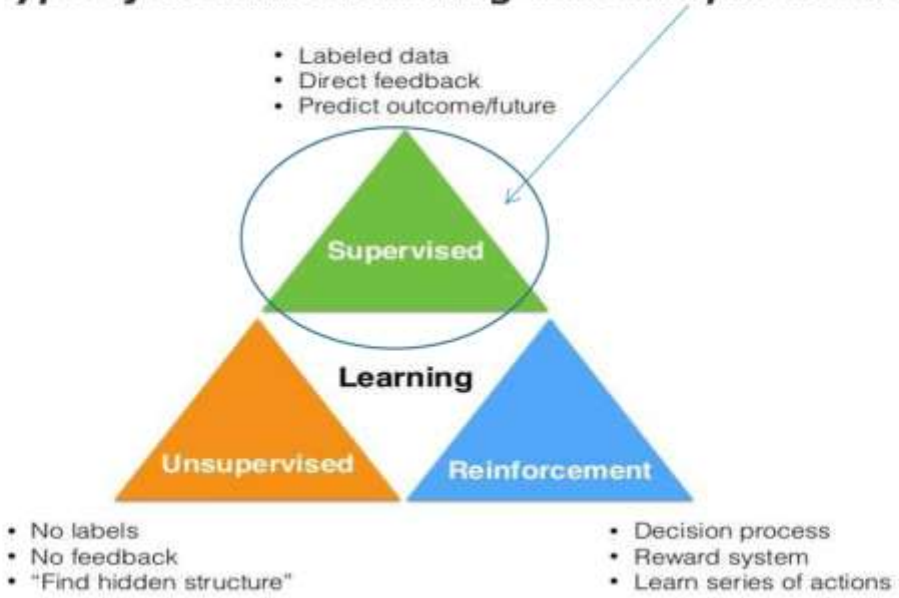
Image

Do you mind if I maybe come over? I'm sorry I don't know if it's going to get better soon

Text

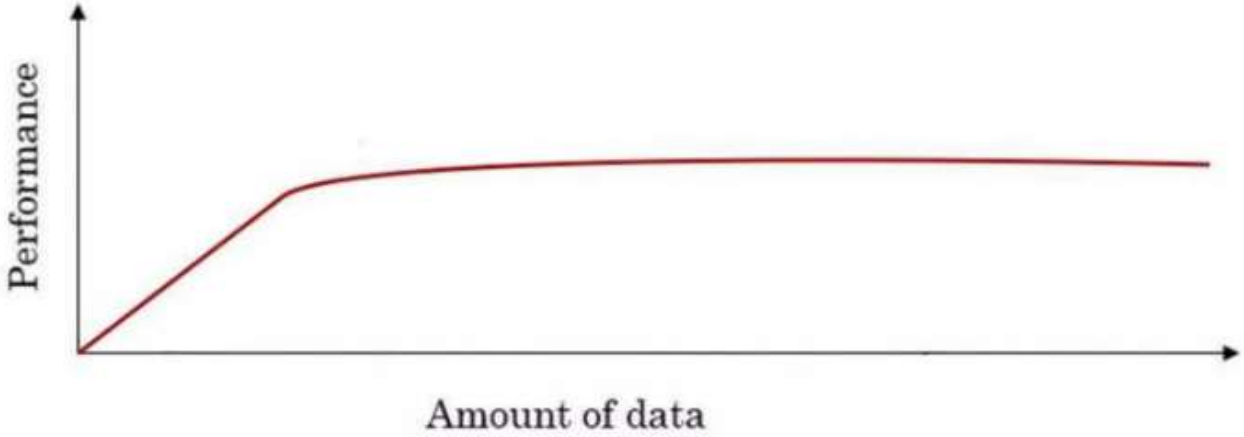
Supervised Learning with Neural Networks

- There's been a lot of hype about neural networks and perhaps some of that hype is justified given how well they're working.
- But it turns out that so far almost all the economic value created by a neural networks has been through ***one type of machine learning called supervised learning.***



The Rise of Deep Learning

- If we plot the performance of a traditional learning algorithms such as Support Vector Machine or Logistic Regression as a function we will get the following curve:



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The Rise of Deep Learning

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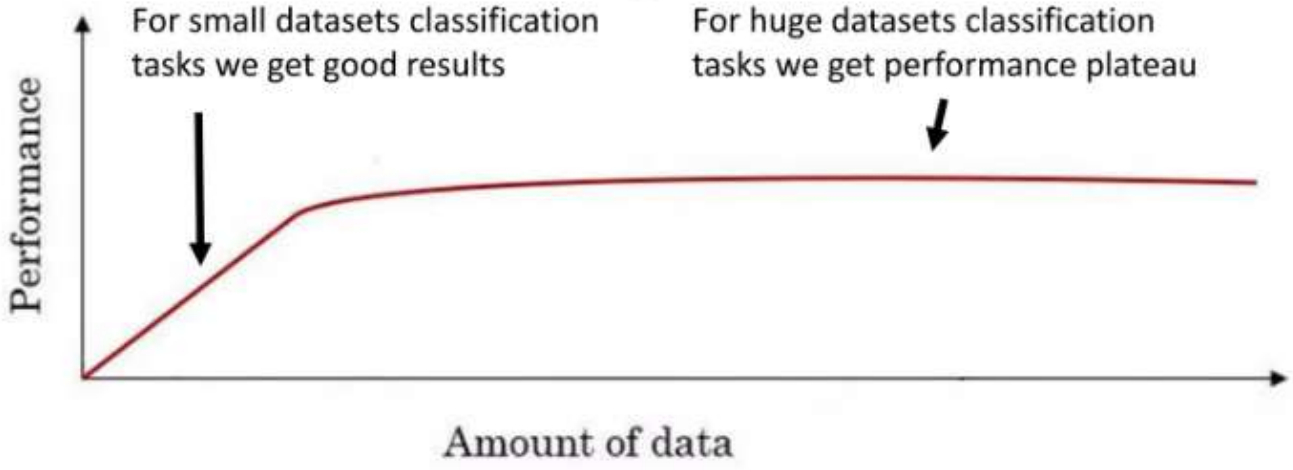

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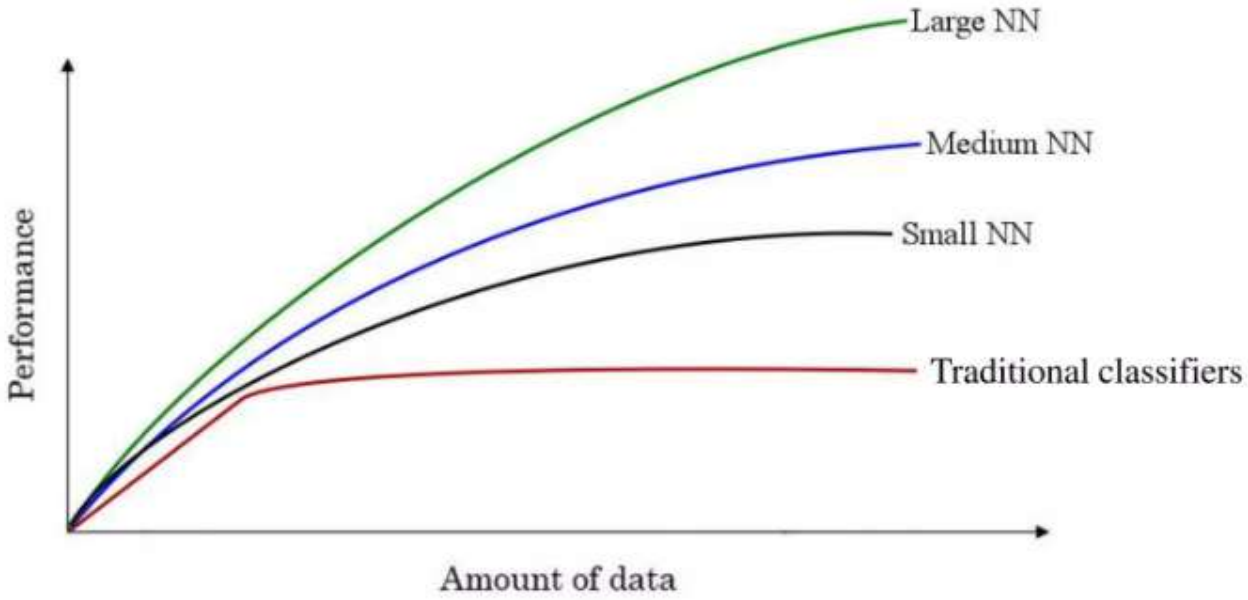
The Rise of Deep Learning

- How to overcome performance plateau problem?



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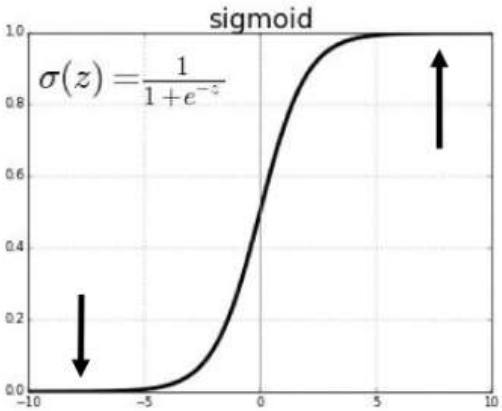
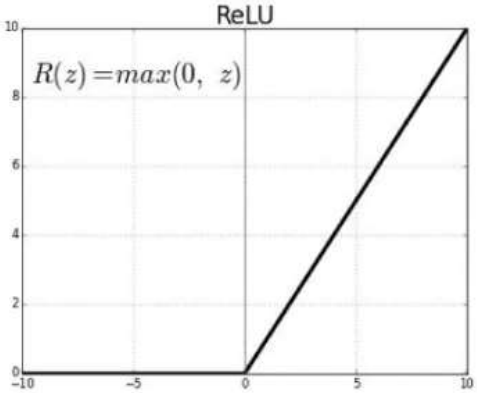
- Let's look in more detail why switching from sigmoid to ReLU function did so well on NN performace.



- One of the problems of using sigmoid functions in machine learning arrises in these regions.
- Here the slope of the function with a gradient is nearly zero that is learning becomes really slow when gradient descent is implemented.
- The gradient is zero and the parameters change very slowly yielding to very slow learning.



- For the Rectified Linear Unit function, the gradient is equal to 1 for all positive values of input.
- The gradient is much less likely to gradually shrink to 0, and the slope of the line on the left is 0.
- At the end, just by switching from the sigmoid function to the RELU function has made an algorithm called gradient descent to work much faster.

Why is deep learning taking off?

1.Data:

- For small data NN can perform as Linear regression or SVM (Support vector machine)
- For big data a small NN is better that SVM
- For big data a big NN is better that a medium NN is better that small NN.
- Hopefully we have a lot of data because the world is using the computer a little bit more: Mobiles, IOT (Internet of things)

2.Computation:

GPUs Powerful CPUs Distributed computing ASICs

3.Algorithm: Creative algorithms has appeared that changed the way NN works.

- For example using RELU function is so much better than using SIGMOID function in training a NN because it helps with the vanishing gradient problem.

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THANK YOU